

ADITYA GOSWAMI

Full-Stack Web Developer

CONTACT

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Full-Stack Web Developer building web applications with React.js, JavaScript, Node.js, Express.js, HTML5 and Tailwind CSS, with MongoDB and MySQL databases. Built StudyRewind, a full-stack application with a FastAPI and PostgreSQL backend, JWT authentication and RAG-style semantic search. Additional practical knowledge of prompt engineering, Retrieval-Augmented Generation (RAG), vector databases, embeddings and AI-assisted development.

SKILLS

Programming Languages: JavaScript, C++, Python, C

Frontend Development: React.js, HTML5, Tailwind CSS

Backend Development: Node.js, Express.js, FastAPI, JWT Authentication, SQLAlchemy 2.0, Alembic

Databases: MongoDB, MySQL, PostgreSQL, pgvector

AI / LLM: Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Databases, Embeddings, Semantic Search, Faster-Whisper, Sentence Transformers

AI-Assisted Development: Strong working knowledge of AI tools such as Claude, Codex, GitHub Copilot and Antigravity, with the ability to use them efficiently and effectively across the development workflow

Computer Science: Data Structures and Algorithms, Object-Oriented Programming (OOP), Computer Networks, DBMS, Software Engineering

EDUCATION

Bachelor of Technology, Computer Science and Business System | 2024 - 2028

Techno International New Town, Kolkata, India (Affiliated with MAKAUT - Maulana Abul Kalam Azad University of Technology)
Currently in 3rd Year, 5th Semester

Class XII | 2024

Sri Chaitanya Techno School

PROJECTS

StudyRewind — AI-Powered Educational Search Assistant

GitHub: <https://github.com/goswamiaditya112-cpu/studyrewinds> | **Video Demo:** <https://youtu.be/5xIW6lkczRc>

Built a full-stack web application that ingests YouTube lecture playlists and PDF study materials, transcribes and embeds content locally using AI, and provides semantic search with timestamp and page provenance.

Backend: FastAPI (Python), PostgreSQL, pgvector, SQLAlchemy 2.0, Alembic, Argon2id authentication, JWT authorization

AI/ML: faster-whisper (local speech transcription), sentence-transformers (paraphrase-multilingual-MiniLM-L12-v2, 384-d embeddings), pgvector cosine similarity search, yt-dlp, youtube-transcript-api, pypdf

Frontend: React 18, TypeScript, Vite

Key engineering work:

- Implemented a three-step transcript extraction pipeline: YouTube captions → yt-dlp subtitle download → local Whisper AI fallback
- Built a local embedding pipeline generating 384-dimensional multilingual vectors (Hindi + English) with batch processing, idempotency, and validation
- Implemented strict user data isolation with IDOR prevention and UUID-based ownership chains across all API endpoints
- Wrote 184 passing automated tests covering authentication, subject management, transcription, PDF processing, semantic search, and security boundaries
- Resolved a Windows file-locking bug in temporary audio cleanup using bounded retry with read-only attribute handling

Portfolio — macOS-Inspired Interactive Developer Portfolio

HTML5 • CSS3 • JavaScript • AI Integration (Planned)

GitHub: https://github.com/goswamiaditya112-cpu/ADITYA-GOSWAMI_PORTFOLIO

Live Portfolio: [Visit my Portfolio \(https://goswamiaditya112-cpu.github.io/ADITYA-GOSWAMI_PORTFOLIO/\)](https://goswamiaditya112-cpu.github.io/ADITYA-GOSWAMI_PORTFOLIO/)

Built an interactive macOS-inspired personal portfolio that presents my profile, projects, skills, experience, education, certificates, resume, and contact information through a desktop-style interface. Implemented a responsive desktop environment with custom wallpaper, menu bar, desktop folders, application Dock, draggable portfolio windows, window controls, window stacking, and responsive layouts using HTML, CSS, and JavaScript.

Future AI Integration: Planning to build a personal, portfolio-grounded AI chatbot that will use my portfolio data to answer questions about my skills, projects, experience, education, and technical background. I also plan to add a job-description matching feature where hiring managers can paste a job description and receive a structured comparison against my portfolio projects and skills. The chatbot architecture, model selection, and AI implementation are currently part of my learning roadmap and are planned for a future version.